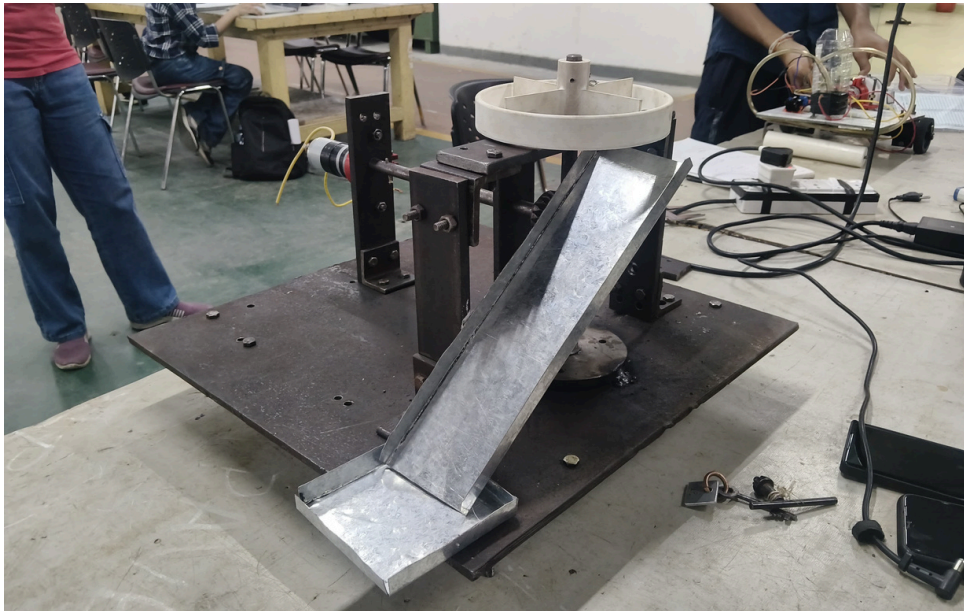




TA 212 LAB (M.E)

2ND SEM. 2025-2026

Automated Medicine Dispenser



STUDENT NAME:-

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3. Paveet Singh Saluja
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6. Manish Gurjar
7. M Rohith Kumar

ROLL NO:-

- (240665)
- (240655)
- (240742)
- (240720)
- (240515)
- (240621)
- (240671)

PROJECT OVERVIEW

Project Objective: The primary objective of this project is to develop an automated smart medicine dispenser using Arduino, sensors, and motors to ensure timely and accurate medication intake for patients, particularly the elderly.

Manufacturing Summary: The system consists of 15 distinct manufactured parts—including mild steel supports, 3D-printed holders, and bevel gears—fabricated using processes such as cutting, drilling, milling, and lathe work .

Cost Estimation: The total project cost is estimated at **Rs. 15939.10**, which includes the cumulative expenses for raw materials, machinery usage, electronic components, and labor.

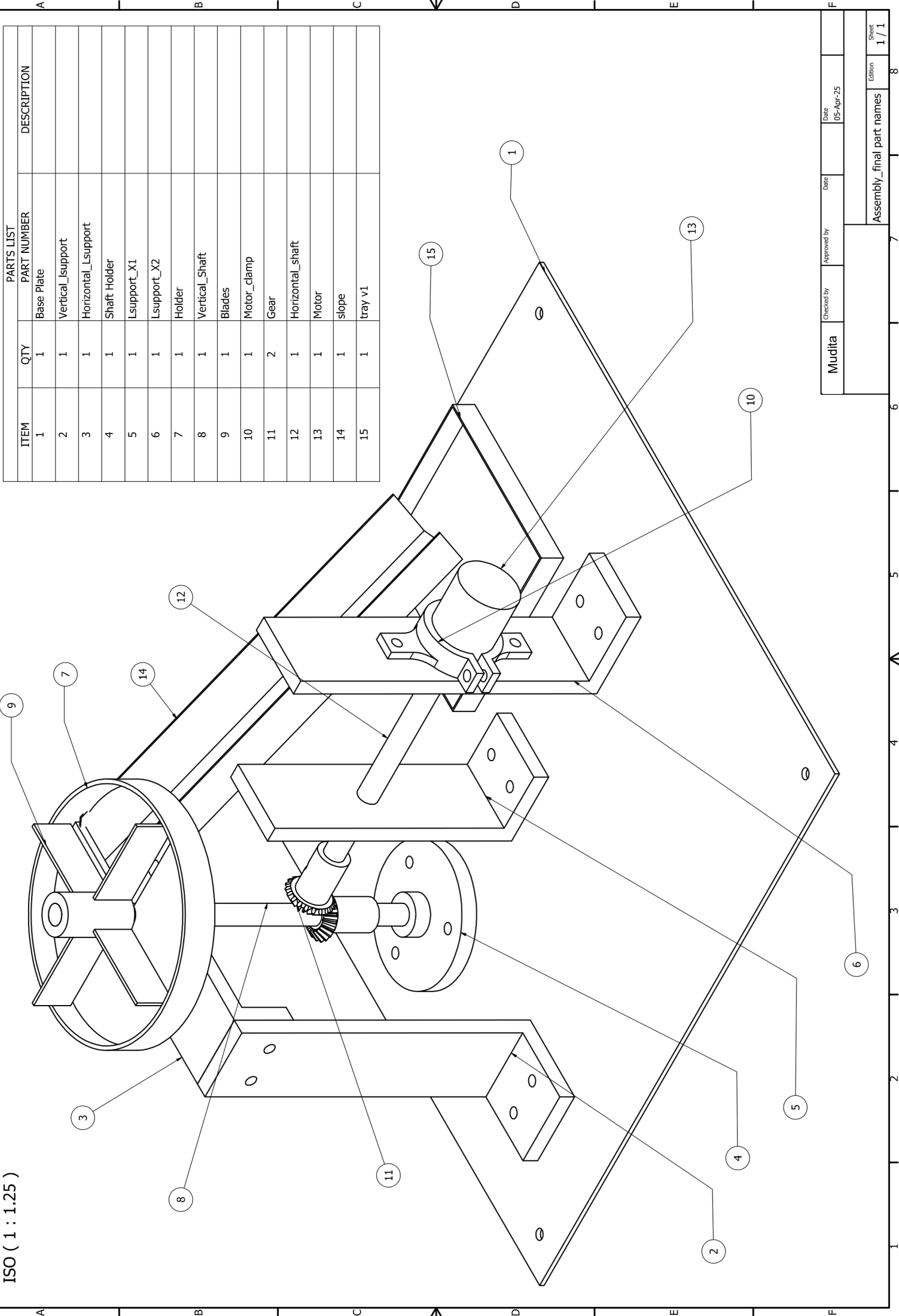
Future Improvements: Suggested improvements include scaling the prototype for multi-user hospital environments and integrating IoT capabilities for remote monitoring and caregiver alerts.

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ISO (1 : 1.25)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Base Plate	A
2	1	Vertical_Lsupport	
3	1	Horizontal_Lsupport	
4	1	Shaft Holder	
5	1	Lsupport_X1	
6	1	Lsupport_X2	
7	1	Holder	
8	1	Vertical_Shaft	B
9	1	Blades	
10	1	Motor_clamp	
11	2	Gear	
12	1	Horizontal_shaft	
13	1	Motor	C
14	1	slope	
15	1	tray v1	



Checked by	Approved by	Date	Date	05-Apr-25
Mudita				
Assembly_final part names				Sheet 1 / 1

Motor Torque Requirements

$$I_{\text{driven}} = I_{\text{blades}} + I_{\text{rod2}} + I_{\text{bevelgear}}$$

Since,

$$I_{\text{rod1}} + I_{\text{rod2}} = 0$$

$$I_{\text{driven}} = I_{\text{blades}} + I_{\text{bevelgear}}$$

$$I_{\text{driven}} = 5.63 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_{\text{driving}} = 1.2 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_{\text{total}} = I_{\text{driven}} + I_{\text{driving}} = 5.63 \times 10^{-3} + 1.2 \times 10^{-3} = 6.83 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$\text{Torque, } M = I_{\text{total}} \times \alpha$$

$$\text{Given } \alpha = \pi/2 \text{ rad/s}^2,$$

$$M = 6.83 \times 10^{-3} \times \pi/2 = 1.07 \times 10^{-2} \text{ N}\cdot\text{m}$$

Taking into account inefficiency (Efficiency=0.5), friction at axle (factor of 2), FOS (factor of 2),

Total factor = 8

$$\text{Required motor torque} = M_{\text{motor}} = 8 \times 1.07 \times 10^{-2} = 0.0856 \text{ N}\cdot\text{m}$$

$$\text{Torque provided by motor} = 1 \text{ N}\cdot\text{m}$$

Therefore a single motor is enough to drive the blades.

$$\text{Required Power} = 0.0856 \times \pi = 0.2687 \text{ Watts}$$

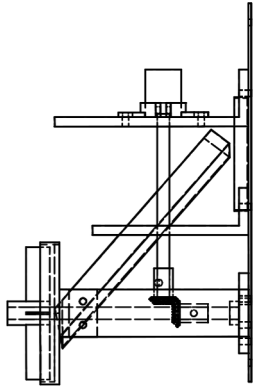
PARTS LIST

Part Name	Quantity	Dimensions in mm	Material	Machining Processes used
Base Plate	1	400*400*3	Mild Steel	Cutting, Drilling
Vertical L Support	1	200*60*50	Mild Steel	Cutting, Drilling
Horizontal L Support	2	125*40*50	Mild Steel	Cutting, Drilling
Shaft Holder	1	Ø85, Ø25, h=20	Mild Steel	Cutting, Drilling
L-Support X1	1	165*50*50	Mild Steel	Cutting, Drilling
L-Support X2	1	205*60*50	Mild Steel	Cutting, Drilling
Holder	1	Ø150, h=20	PLA	3D Printing
Vertical Shaft	1	Ø12.7, h=245	Mild Steel	Cutting, Drilling
Blades	1	h=45	Mild Steel	Cutting, Welding
Motor Clamp	1	Ø48, Ø38	PLA	3D Printing
Bevel Gear	2	Ø12.7, h=36.23	Mild Steel	Milling, Lathe
Horizontal Shaft	11	Ø12.7, h=200	Mild Steel	Cutting, Drilling
Slope	1	280*40*35	GI Sheet	Cutting, Welding
Tray	1	120*15	GI Sheet	Cutting

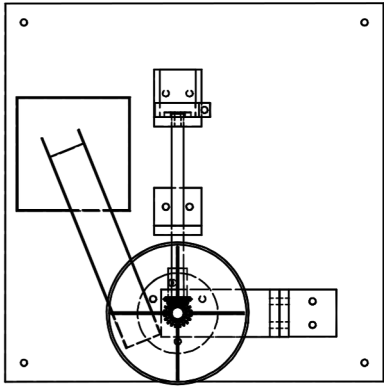
NOTES:

1. PART TO BE POWDER COATED

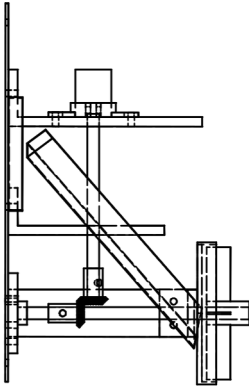
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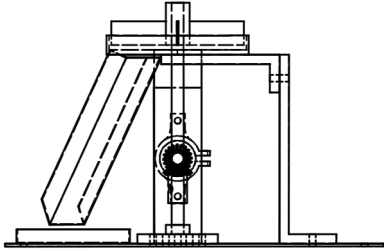
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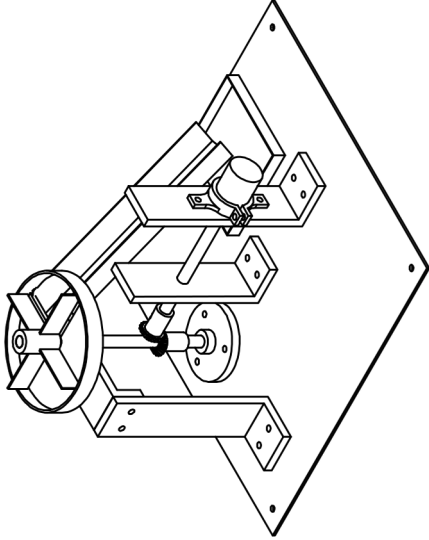
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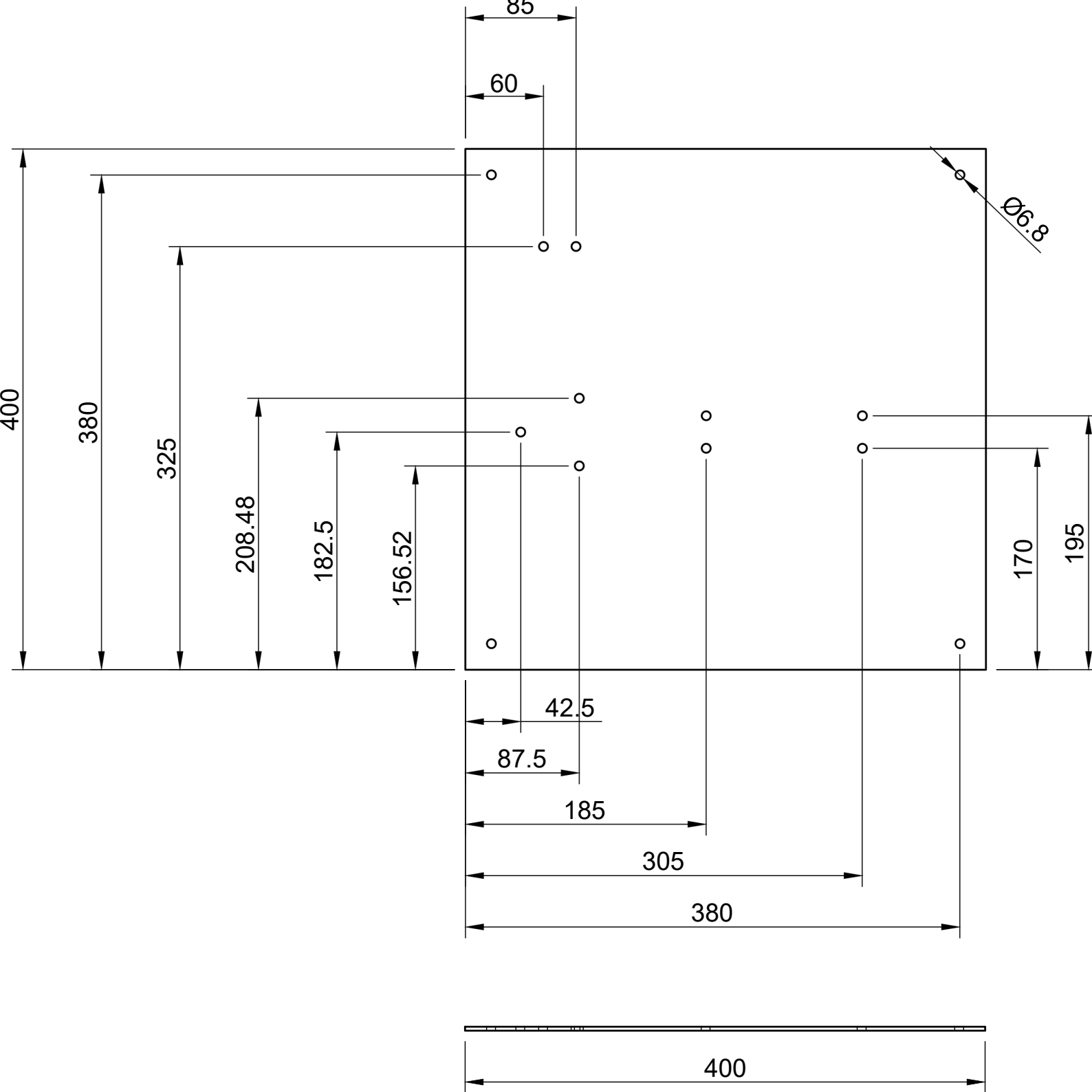


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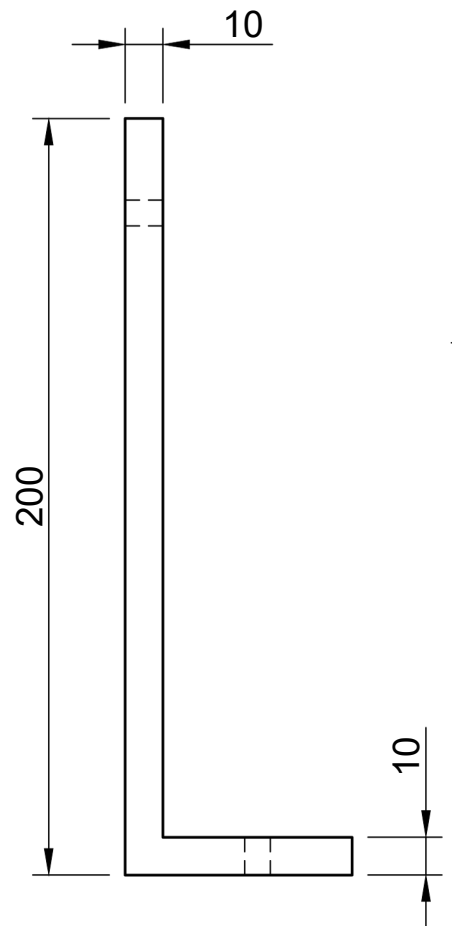
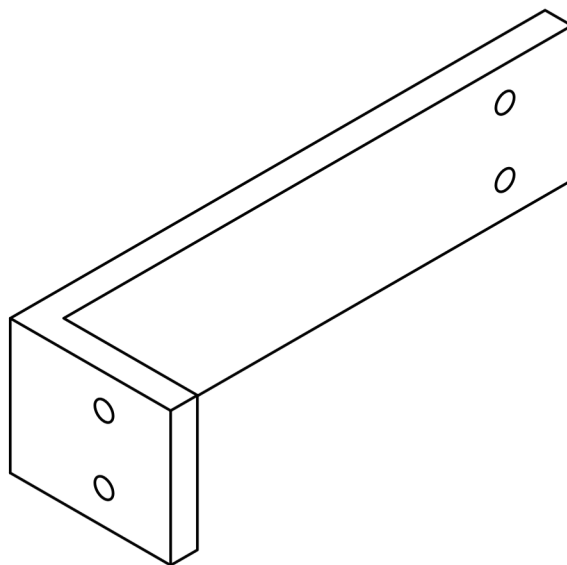
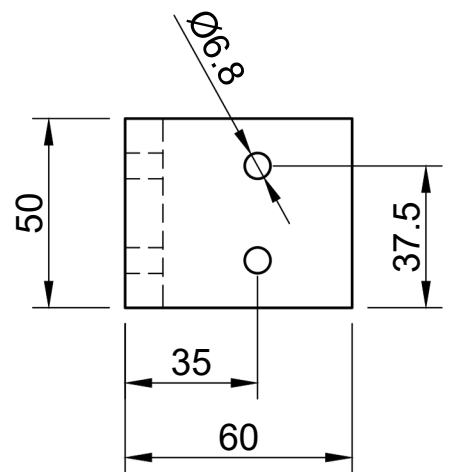
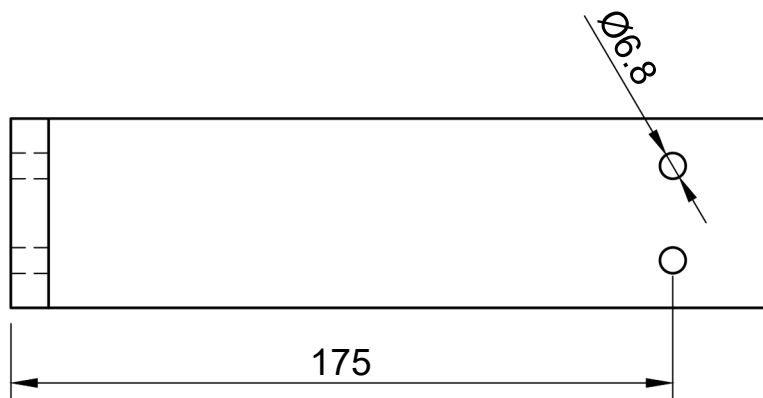


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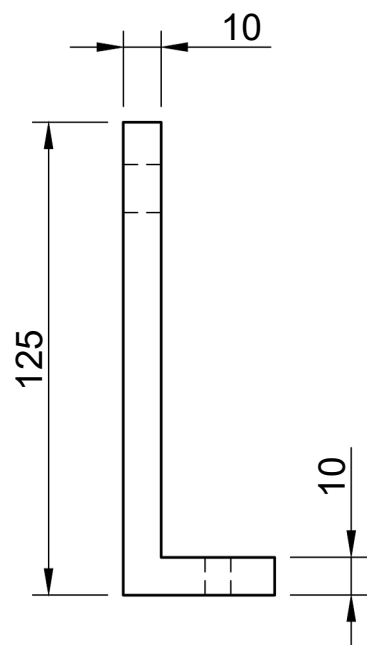
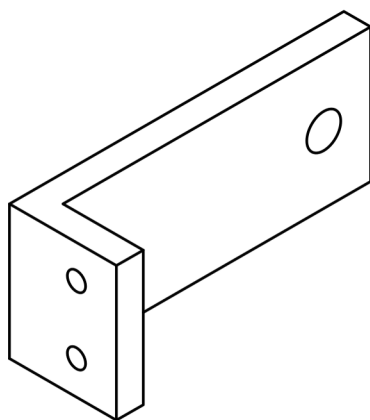
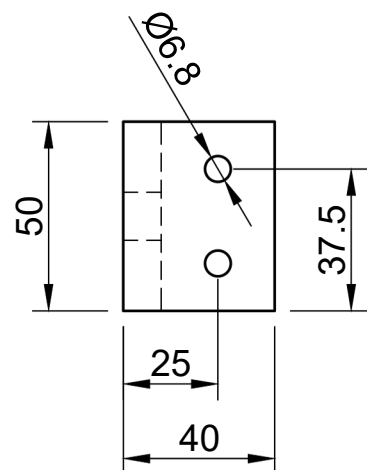
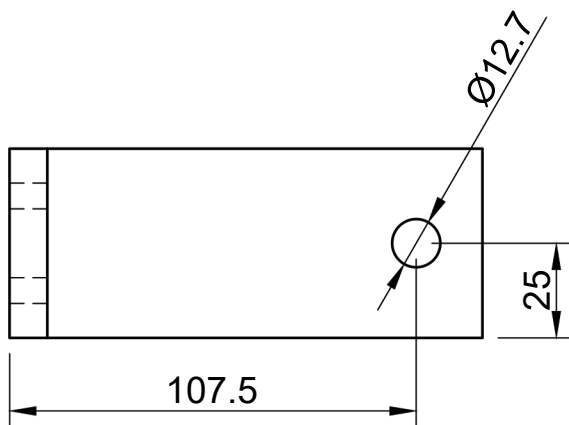




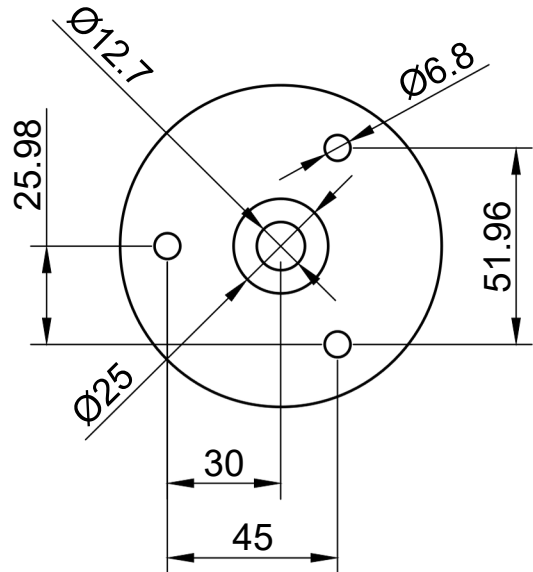
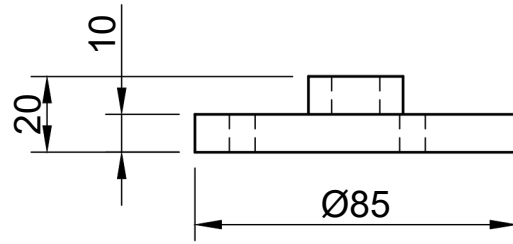
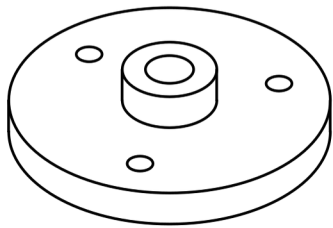
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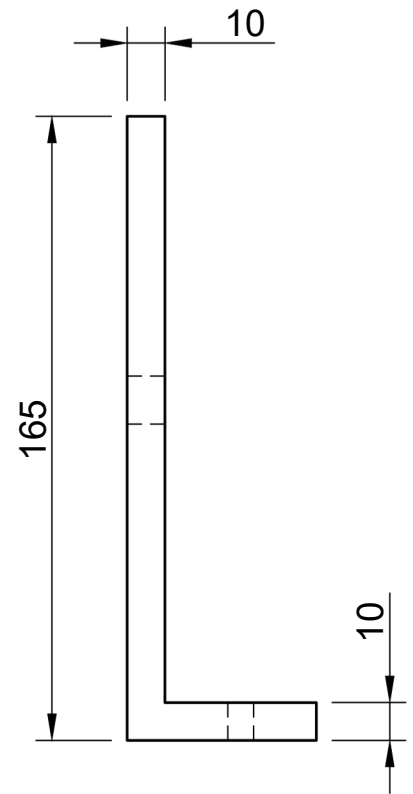
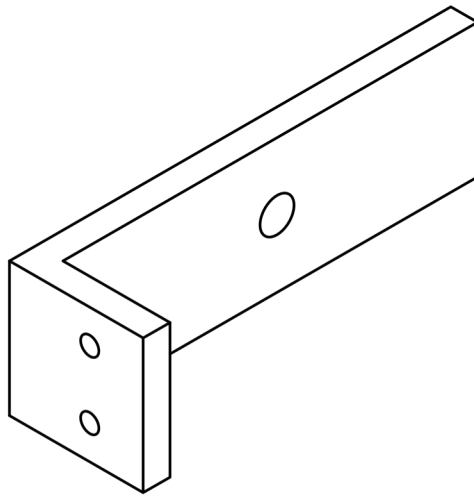
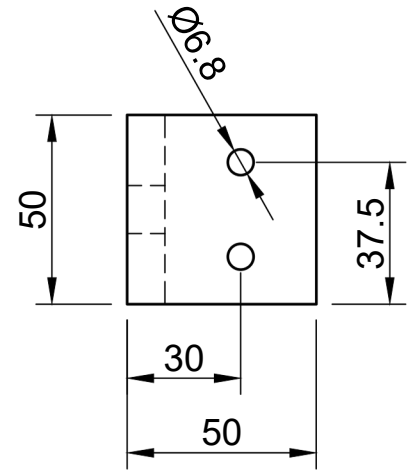
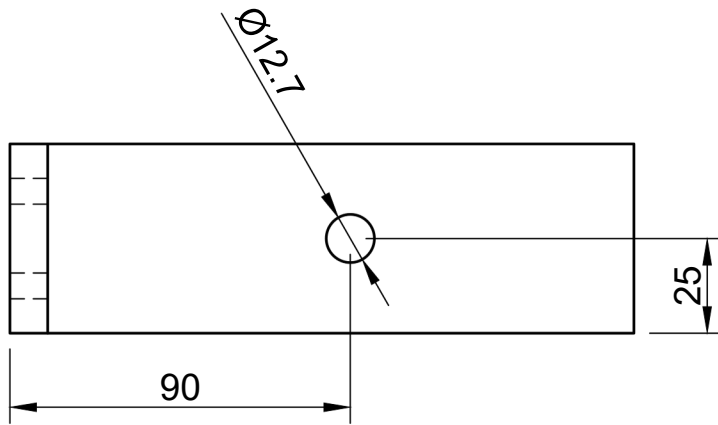
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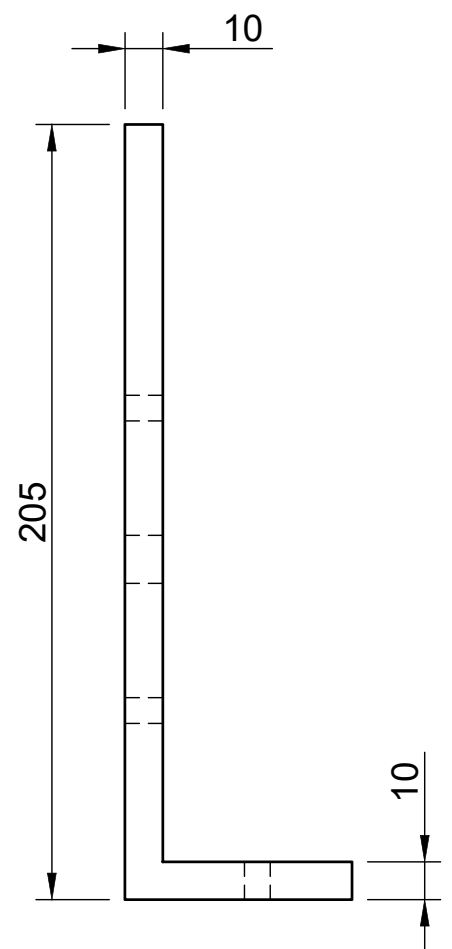
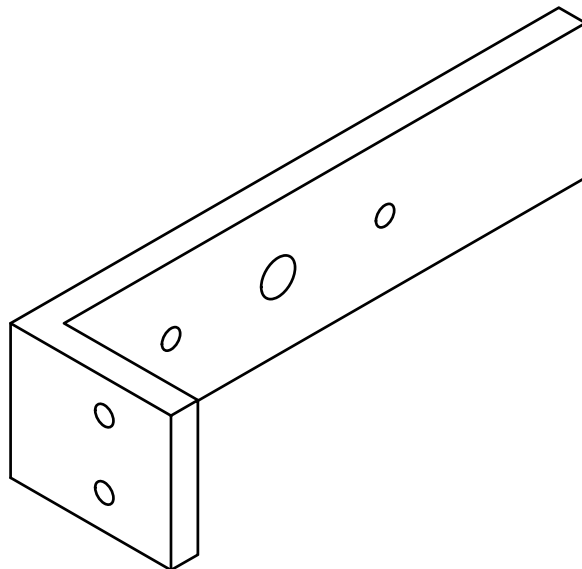
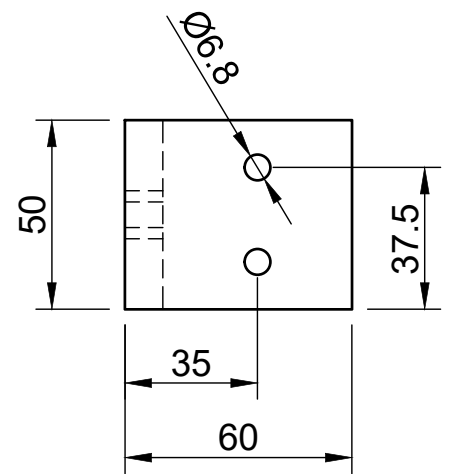
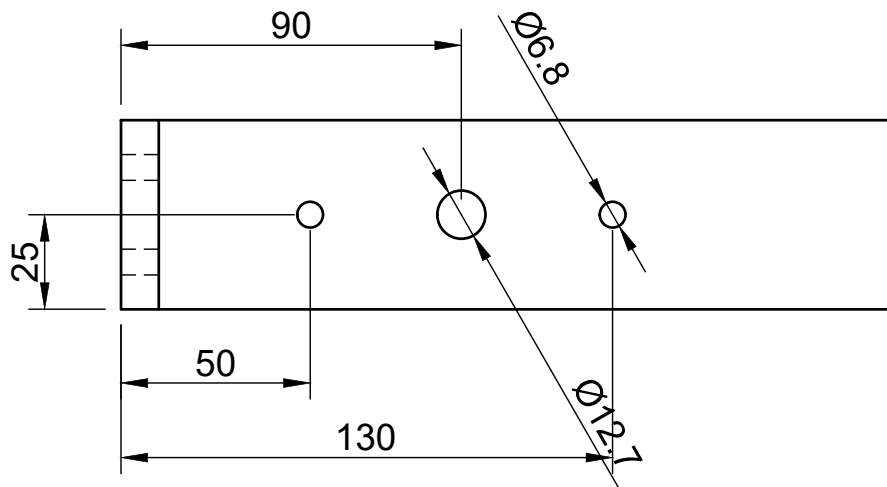
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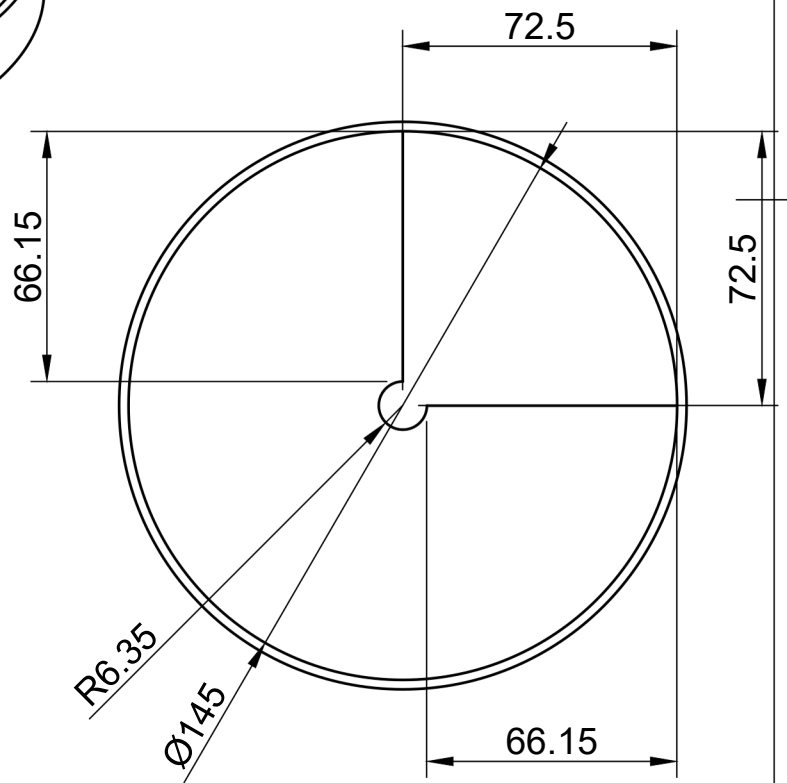
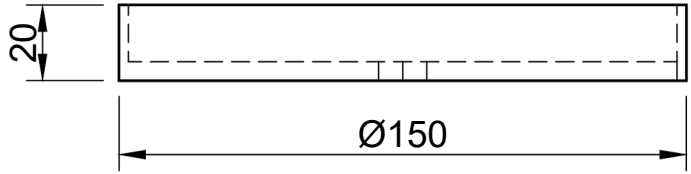
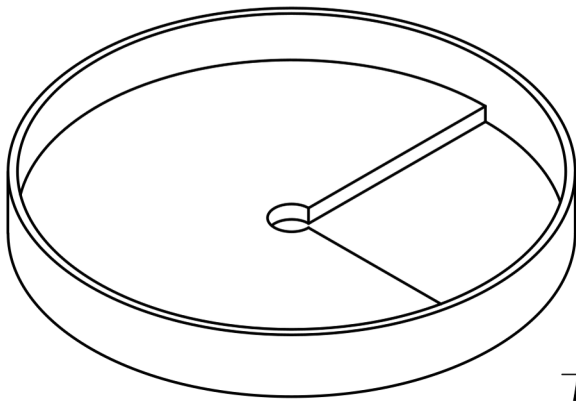
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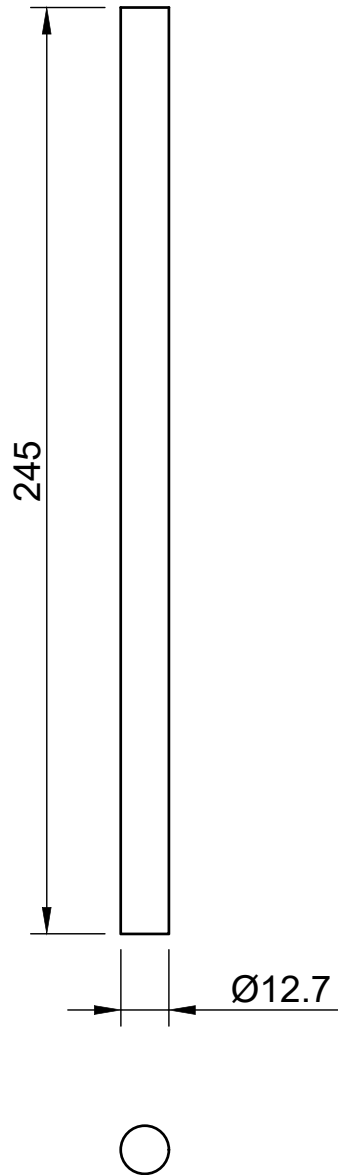
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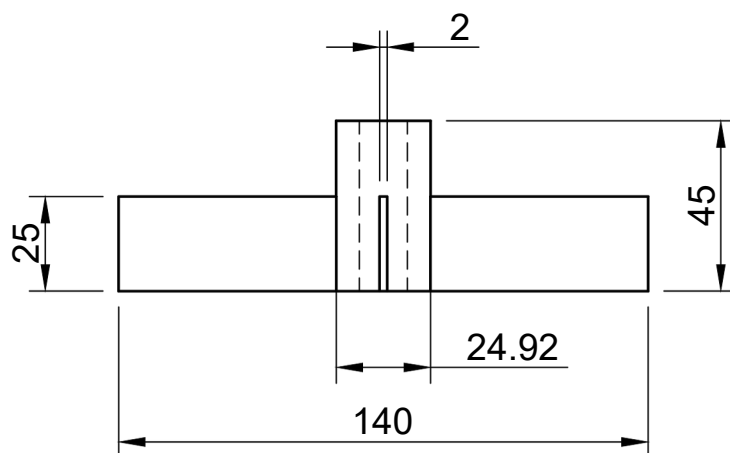
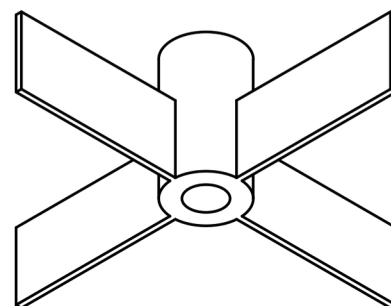
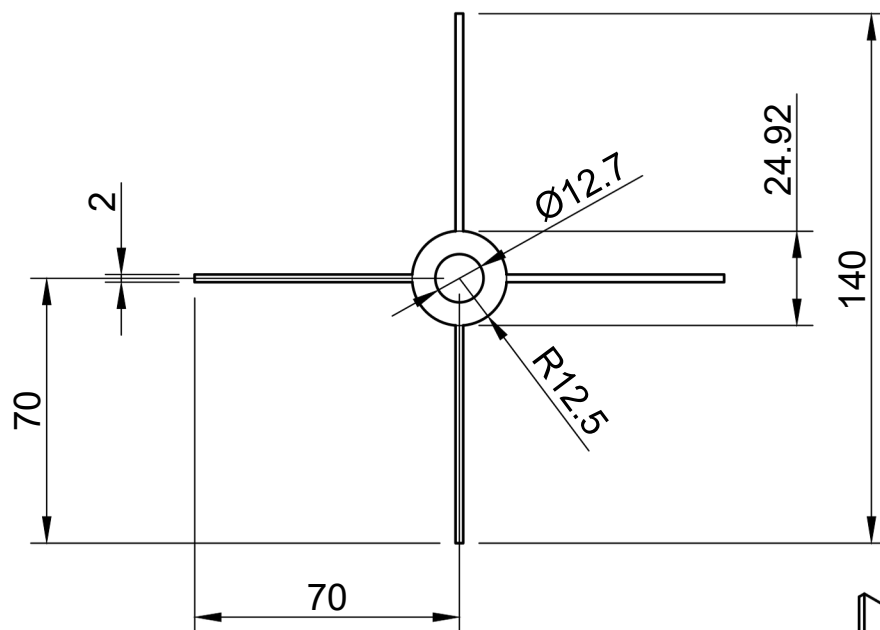
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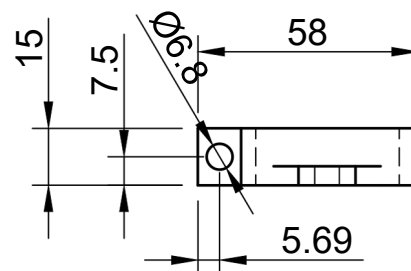
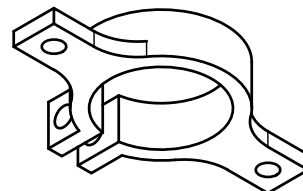
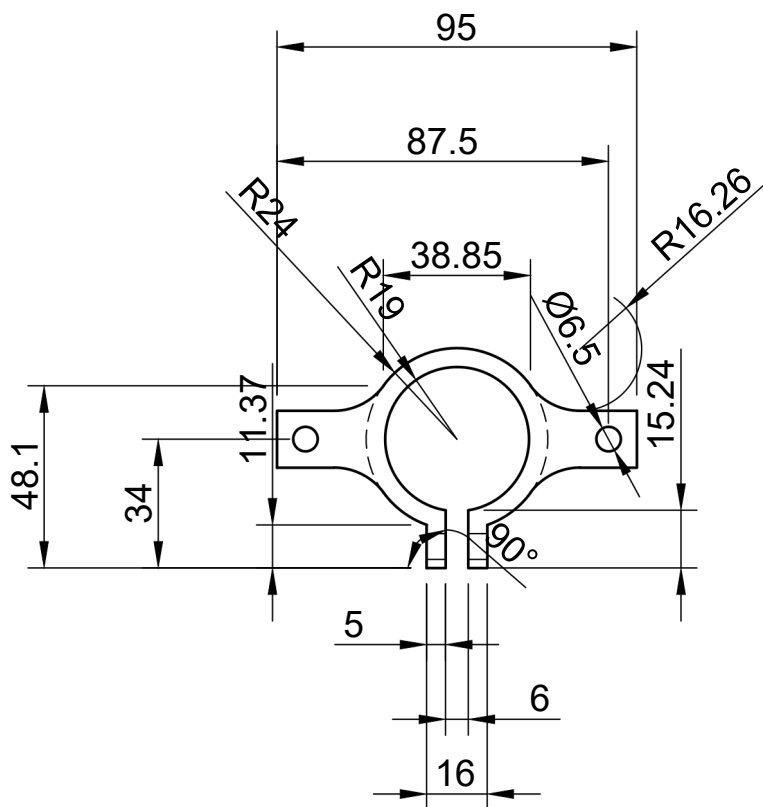
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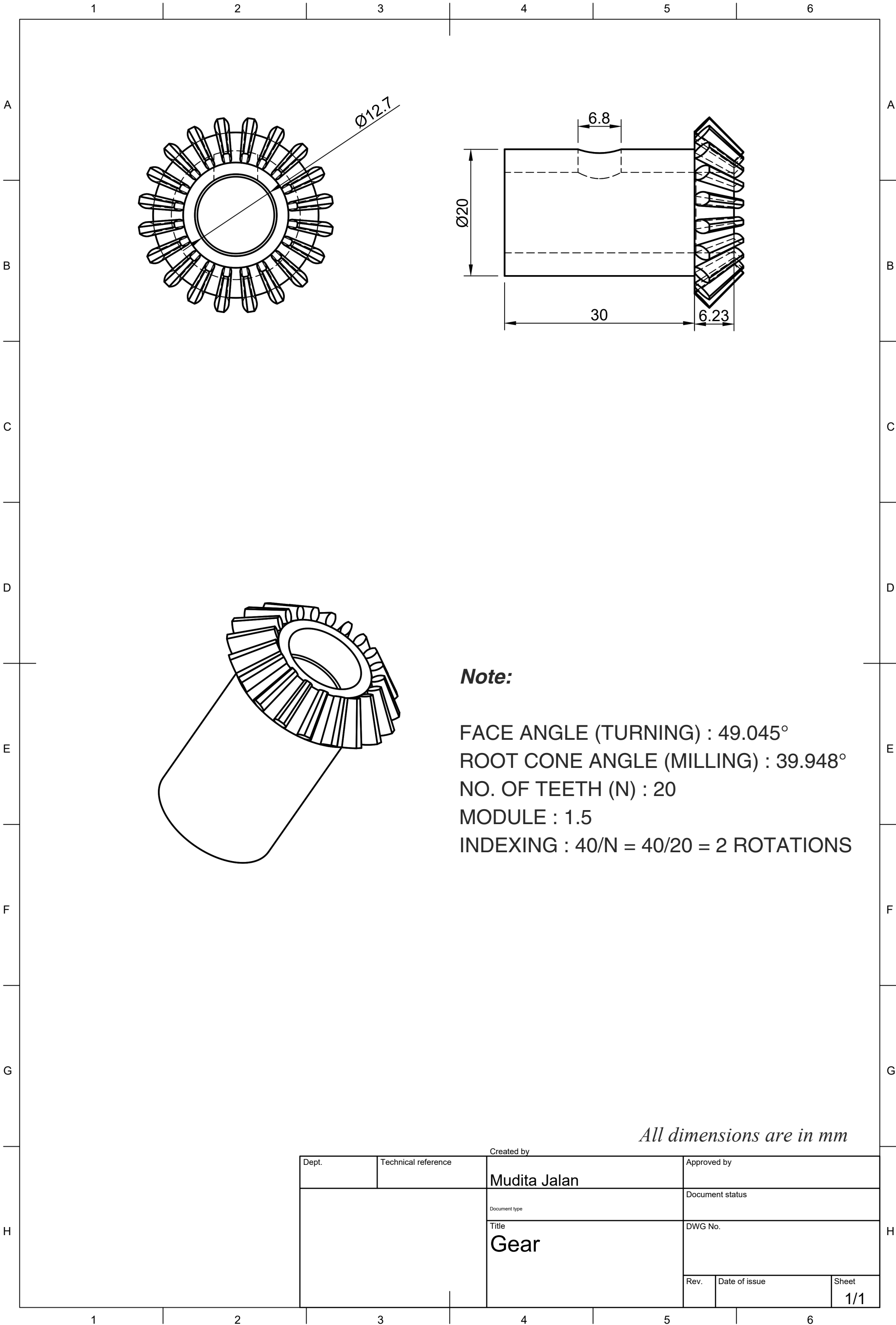
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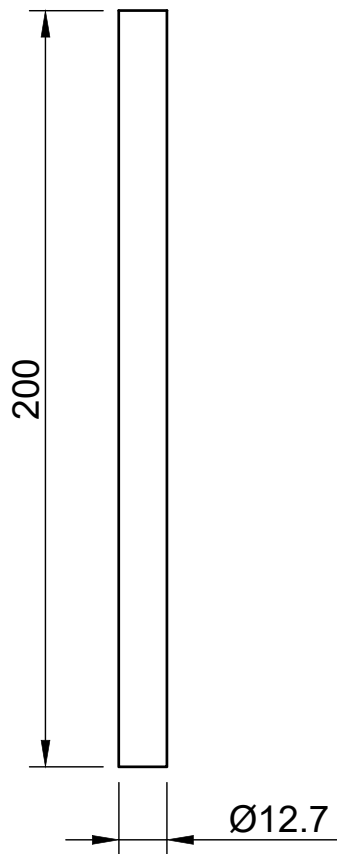


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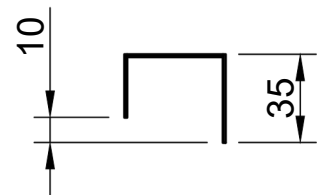
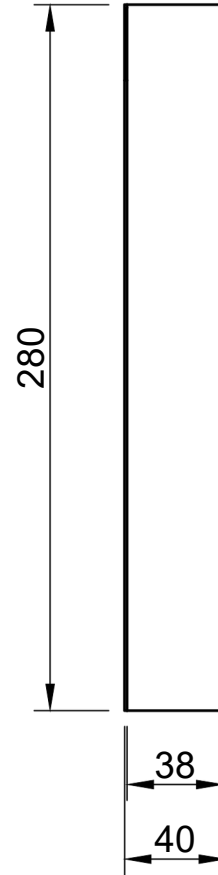
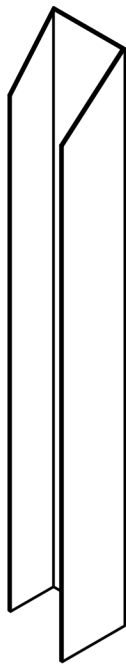
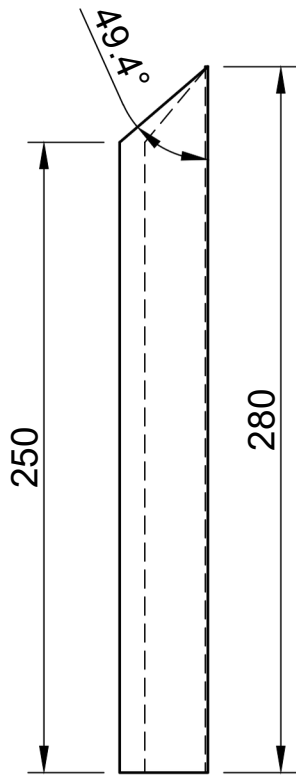


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			Rev.	Date of issue	Sheet 1/1

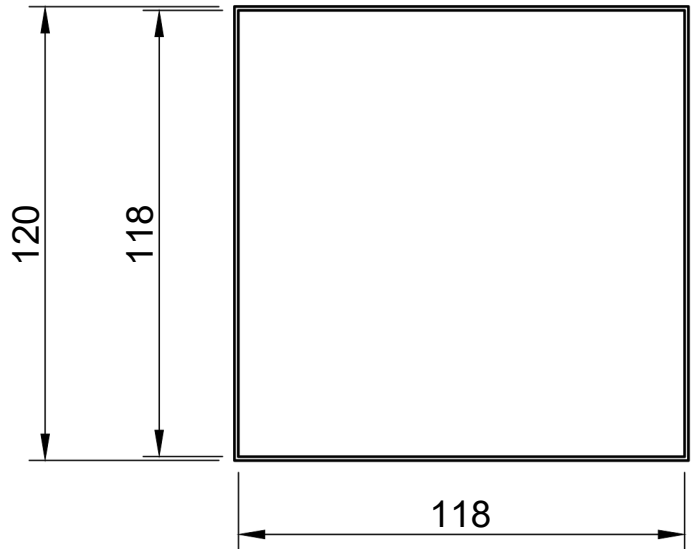
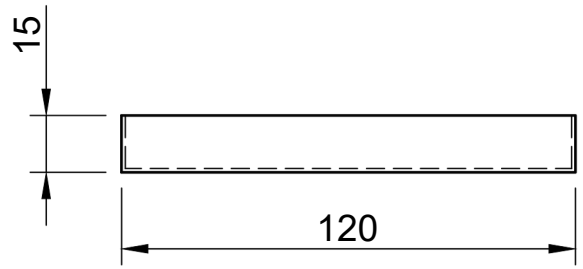
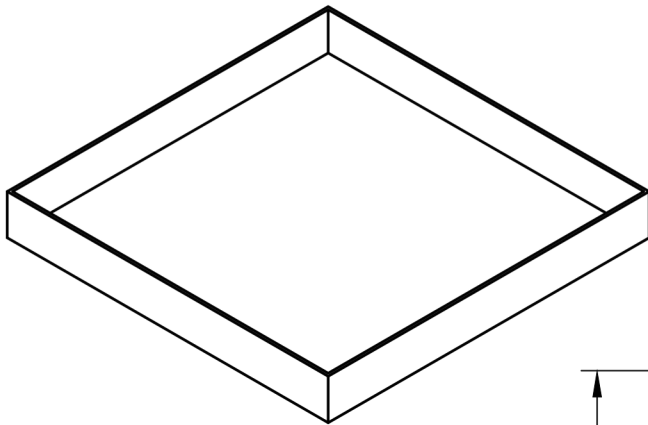




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Straight Bevel Gear Calculation Table

Module = 2, Number of Teeth = 20

S.No.	Item	Symbol	Formula	Example
1	Shaft Angle	Σ		90°
2	Module	m		2
3	Pressure Angle	α		20°
4	Number of Teeth	z_1, z_2		20, 20
5	Pitch Diameter	d	zm	40
6	Pitch Cone Angle	δ_1 δ_2	$\tan^{-1} \left(\frac{\sin \Sigma}{z_2/z_1 + \cos \Sigma} \right)$ $\Sigma - \delta_1$	45° 45°
7	Cone Distance	R_e	$\frac{d}{2 \sin \delta_2}$	28.284
8	Face Width	b	Less than $R_e/3$	9.428
9	Addendum	h_a	$1.00m$	2.00
10	Dedendum	h_f	$1.25m$	2.50
11	Addendum Angle	θ_a	$\tan^{-1}(h_a/R_e)$	4.045°
12	Dedendum Angle	θ_f	$\tan^{-1}(h_f/R_e)$	5.052°
13	Outer Cone Angle	δ_a	$\delta + \theta_a$	49.045°
14	Root Cone Angle	δ_f	$\delta - \theta_f$	39.948°
15	Outside Diameter	d_a	$d + 2h_a \cos \delta$	42.828
16	Pitch Apex to Crown	X	$R_e \cos \delta - h_a \sin \delta$	18.586
17	Axial Face Width	X_b	$\frac{b \cos \delta_a}{\cos \theta_a}$	6.196
18	Inner Outside Diameter	d_i	$d_a - \frac{2b \sin \delta_a}{\cos \theta_a}$	28.552

COST ANALYSIS

MATERIAL(MILD STEEL , ANGLE, ROD, NUT BOLT.)

MATERIAL	TOTAL WEIGHT	RATE	COST (Rs)
MILD STEEL (ANGLE, ROD)	9.93 KG	90 Rs/KG	894
NUT BOLT	0.566 KG	130 Rs/KG	73

MACHINERY COST

MACHINE	RATE	TIME (Hr)	COST (Rs)
3D PRINTING	300 Rs/Hr	5	1500
LATHE	350 Rs/Hr	5	1750
MILLING	450 Rs/Hr	3	1350
DRILLING	100 Rs/Hr	2.5	250

LABOUR COST

TYPE	RATE (Rs/8Hr)	TIME (Hr)	COST (Rs)
SKILLED	850	25	2656
UNSKILLED	650	114	9262

ELECTRONIC COST

WITH SINGLE MOTOR – 2500 Rs

TOTAL

TYPE	COST
RAW MATERIAL	967
MACHINERY	4850
LABOUR	11918
ELECTRONIC	2500
TOTAL	20,235 Rs